

EXHIBIT 18

Exhibit 18: U.S. Patent No. 12,412,924 – EVE 50E

This claim chart shows how each element of the independent claim of U.S. Patent No. 12,412,924 (“the ’924 Patent”) is met by the EVE 50E battery cell (“EVE 50E”), either literally or under the doctrine of equivalents. The materials referenced, and the citations, are merely exemplary and are not exhaustive. LG Energy Solution (“LGES”) reserves the right to rely on different or additional sources and/or different or additional citations to sources that further illustrate infringement. The evidence of infringement provided below is representative in nature and subject to change as discovery commences, including expert discovery. Further, the ALJ has not yet construed certain claim terms of the ’924 Patent. Accordingly, LGES also expressly reserves the right to supplement or amend its claim charts as discovery progresses.

Claims	Exemplary Evidence of Infringement
1[pre]. A separator for a lithium battery comprising:	The EVE 50E is a lithium battery that includes a separator.  As indicated by EVE Energy North America, the EVE 50E is a “lithium-ion rechargeable” battery.



Cylindrical NCM
Cells
(18650/21700)



Prismatic LFP Cells

See <https://www.evebatteryusa.com/lithium-ion> (last accessed June 12, 2026).

CYLINDRICAL NCM CELL SPECIFICATIONS

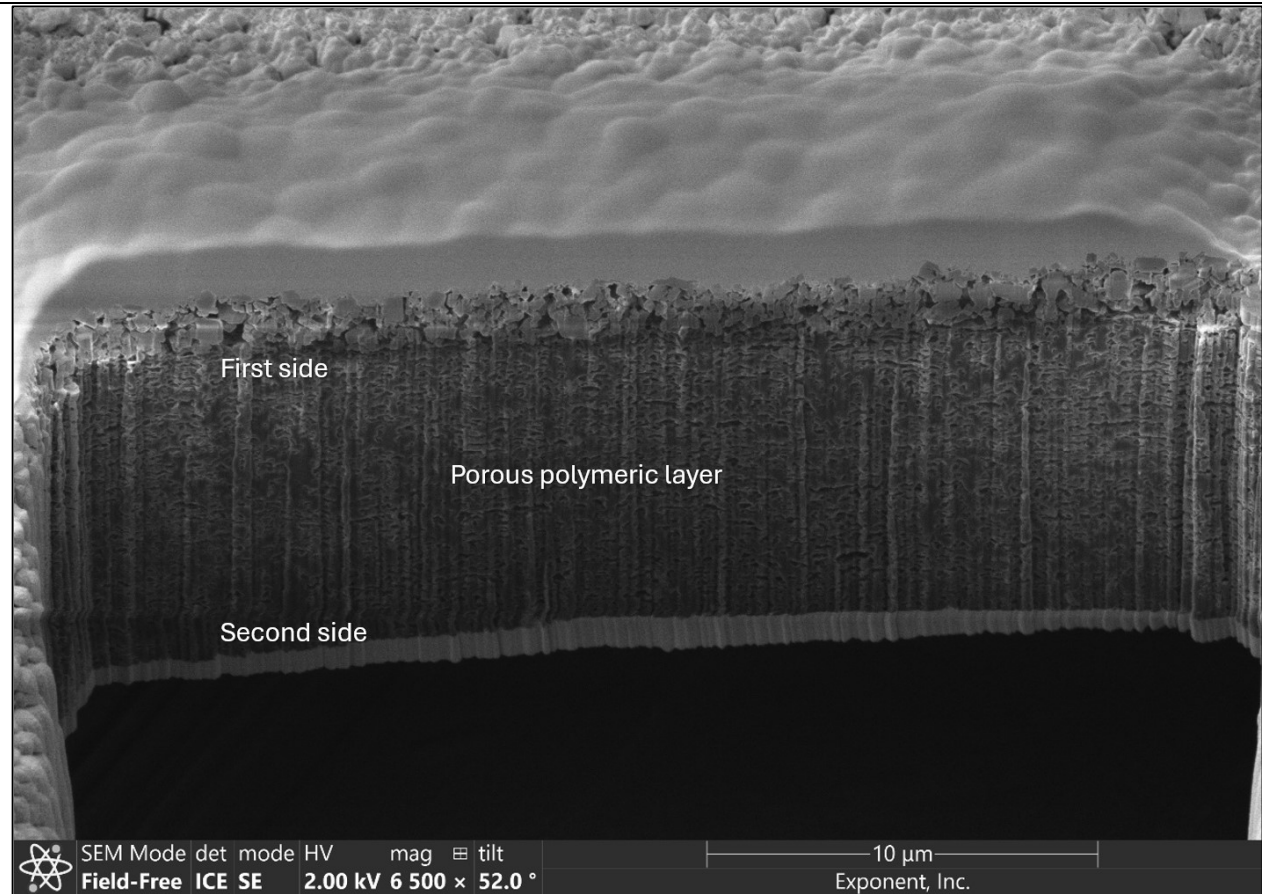


Cell Model	Capacity (Ah)	Nominal Voltage (V)	Max. Cont. Charge/Discharge	Pulse Charge/Discharge	Height (mm)	Diameter (mm)	Weight (g)
50E (21700)	5	3.65	1C/3C		70.3	21.25	69

See <https://www.evebatteryusa.com/18650-and-21700-specifications>. (last accessed June 12, 2026).

As shown below, the EVE 50E includes a separator.

	 Image of the separator extracted from the EVE 50E.
1[a]. a porous polymeric layer having a first side and a second side;	The EVE 50E's separator includes a porous polymeric layer having a first side and a second side. As shown in the cross-section scanning electron microscopy (“SEM”) image below, the separator comprises a porous polymeric layer having a first side and a second side.

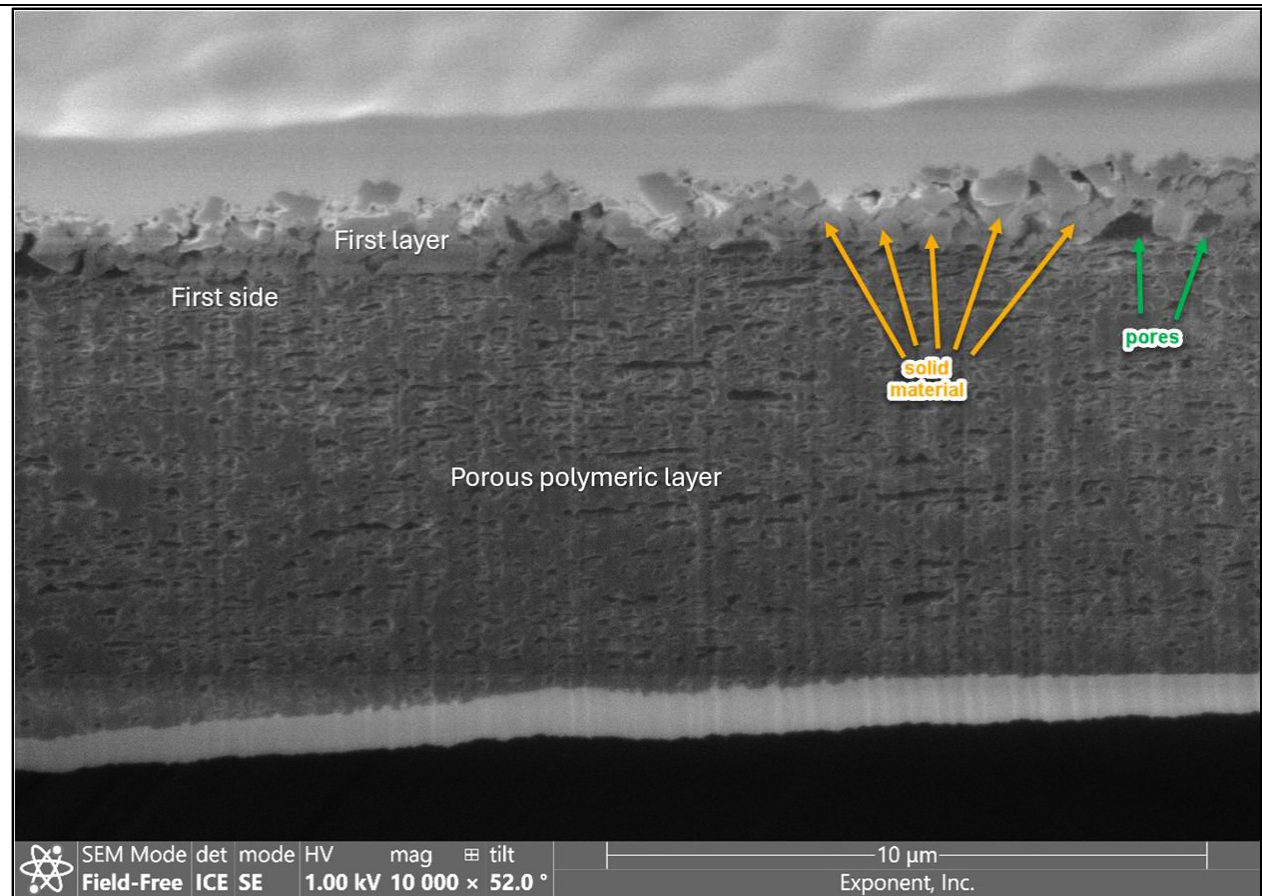


SEM image of the cross section of EVE 50E separator milled by a focused ion beam (“FIB”) (annotations added).

1[b]. a first layer adjacent the first side of the porous polymeric layer, said first layer comprising pores and solid material, wherein

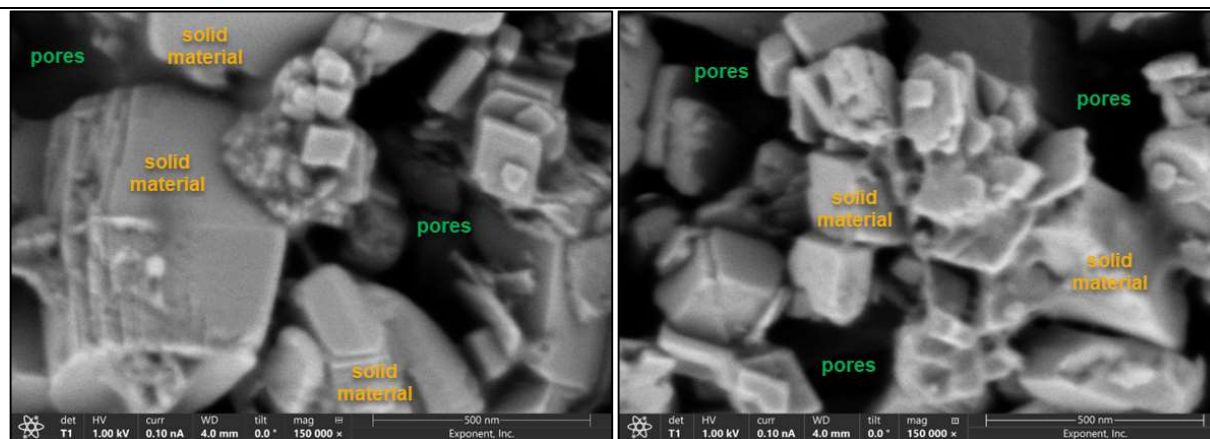
The EVE 50E’s separator includes a first layer adjacent the first side of the porous polymeric layer, said first layer comprising pores and solid material.

As shown in the cross-section SEM image below, the separator shows a first layer adjacent to the first side of the porous polymeric layer. The image further shows that the first layer includes pores (green) and solid material (orange).



SEM image of the FIB cross-section of EVE 50E separator (annotations added).

When viewed from top down, the first layer under SEM also shows pores and solid material.

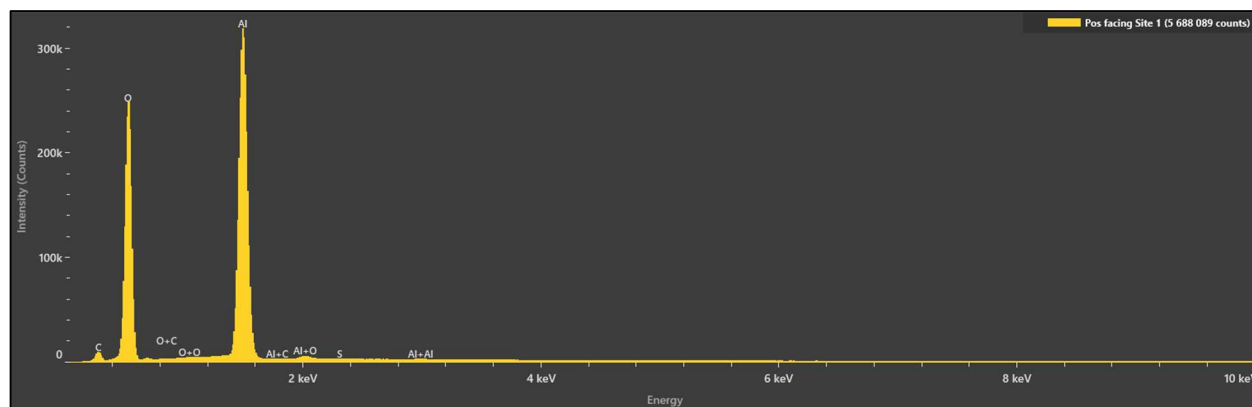


SEM images of the first layer of the EVE 50E separator (annotations added).

1[c]. (a) the solid material comprises inorganic oxide particles and at least one polymer;

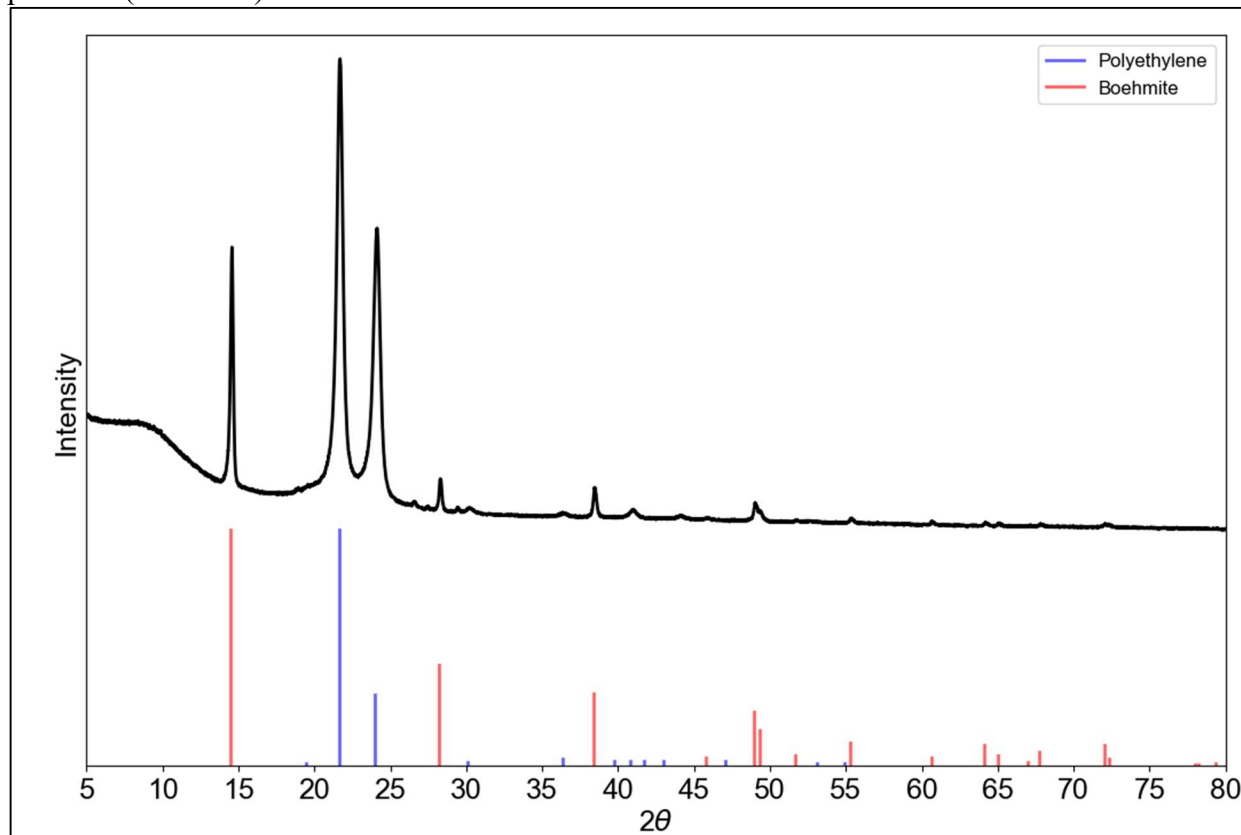
The EVE 50E's separator includes solid material that comprises inorganic oxide particles and at least one polymer.

Scanning electron microscopy combined with energy dispersive spectroscopy (SEM-EDS) analysis of the solid material confirms the presence of an inorganic oxide compound (boehmite) with strong aluminum and oxygen peaks.



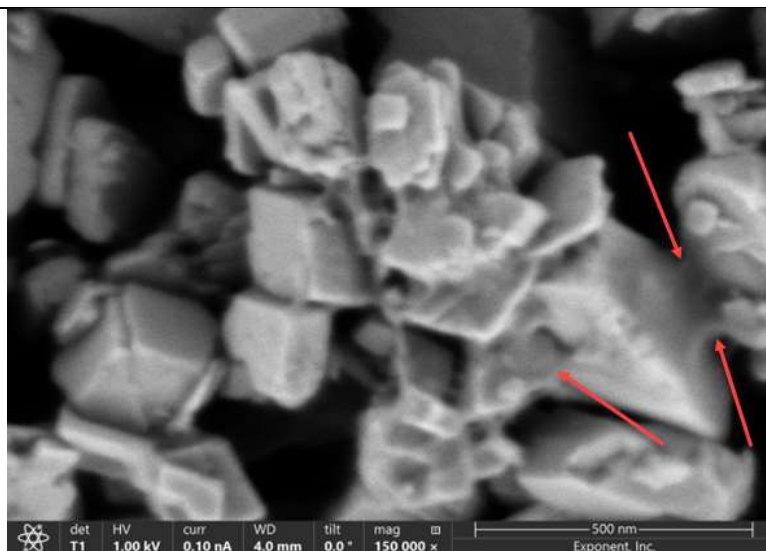
SEM-EDS analysis of the solid material in the first layer of the separator in the EVE 50E.

X-ray diffraction (XRD) analysis further confirms the solid material comprises inorganic oxide particles (boehmite).



XRD analysis of the solid material in the first layer of the separator in the EVE 50E.

Further, the boehmite particles adhere to the porous polymeric layer of the separator, as shown in the cross-section SEM images above. On information and belief, the first layer includes at least one polymer that functions as a binder to allow the boehmite particles to adhere to the surface of the porous polymeric layer. This polymer binder can also be observed, for example, in the surface SEM image below, as indicated by the red arrows.



SEM images of the first layer of the EVE 50E separator (annotations added).

1[d]. (b) a plurality of the inorganic oxide particles in the first layer comprise crystallites having a size in the range of 5 nm to 90 nm;

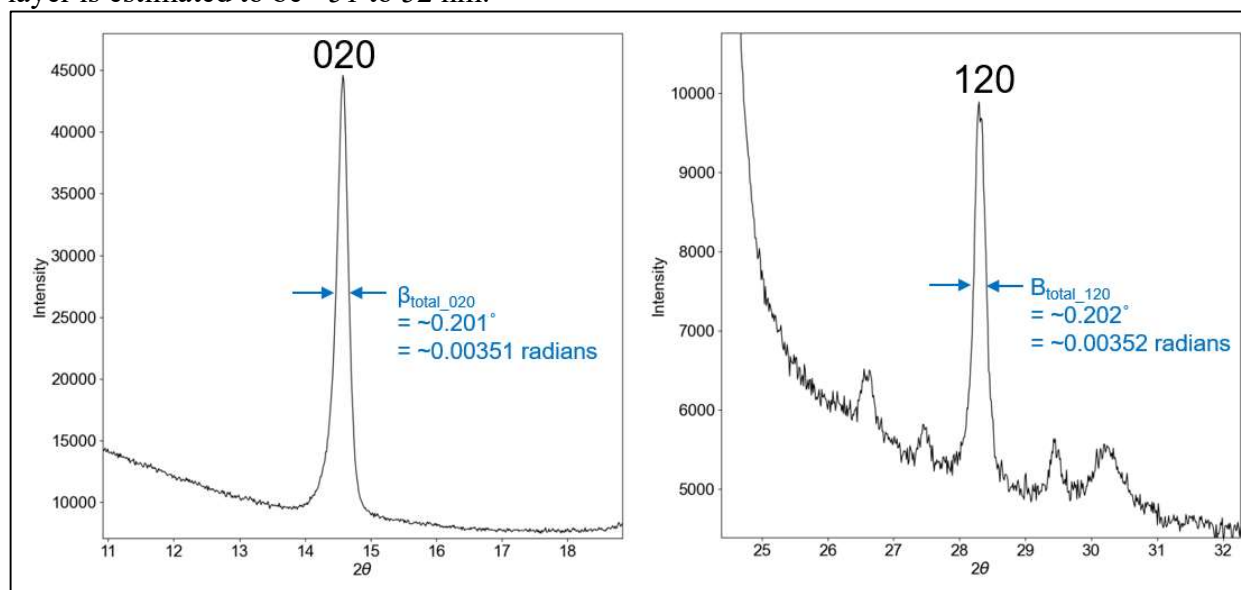
The EVE 50E's separator includes a plurality of the inorganic oxide particles in the first layer that comprise crystallites having a size in the range of 5 nm to 90 nm.

As shown in the top-down SEM images below, a plurality of the boehmite particles in the first layer has a size in the range of 5 nm to 90 nm, *e.g.*, ~60 nm or ~70 nm.



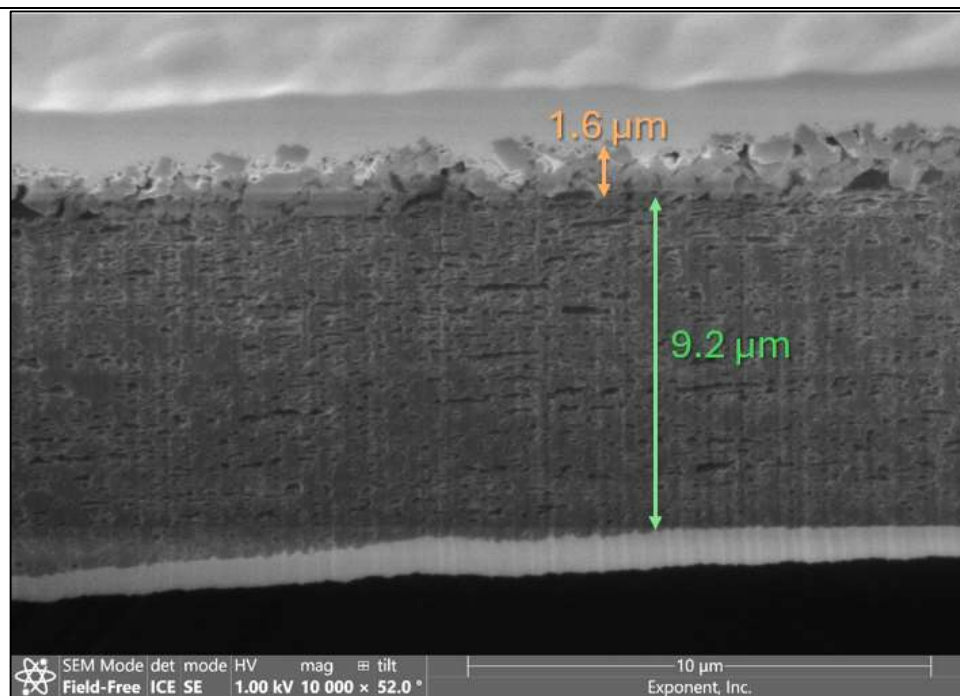
SEM images of the first layer of the EVE 50E separator (annotation added).

Further, using the Scherrer equation, the average crystallite size of the boehmite particles in the first layer is estimated to be ~51 to 52 nm.



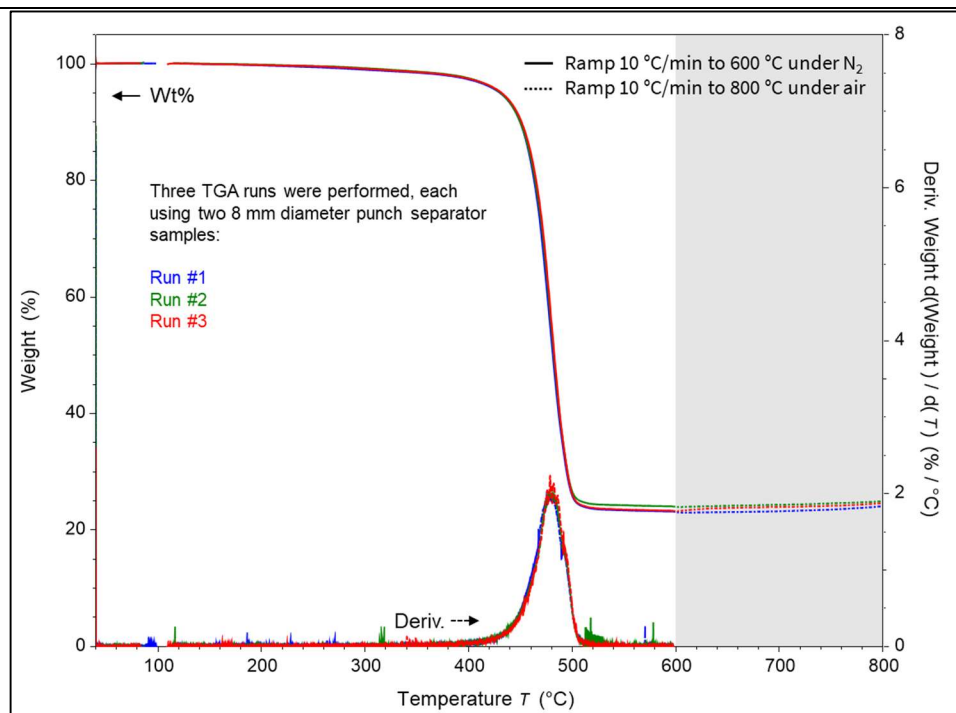
XRD analysis for boehmite in the first layer of the EVE 50E separator.

	Scherrer Equation: $D = \frac{K\lambda}{\beta_{sample} \cos(\theta)}$ D = mean crystallite size K = shape factor, assumed to be 0.9 λ = wavelength of source, 0.154 nm β_{sample} = corrected full width at half maximum* θ = Bragg angle $\beta_{total}^{1.5} = \beta_{sample}^{1.5} + \beta_{inst}^{1.5}$ β_{inst} = 0.00161 rad (same value for both peaks) β_{sample} = ~0.00274 rad (for 020) β_{sample} = ~0.00276 rad (for 120) D_{020} = ~51 nm (using 020 peak at $2\theta = 14.57^\circ$) D_{120} = ~52 nm (using 120 peak at $2\theta = 28.31^\circ$) Accordingly, the mean crystallite size of the boehmite was calculated to be ~51-52 nm using the XRD results for the 020 and 120 peaks and the Scherrer Equation after correcting for the instrument broadening. As such, the first layer comprises crystallites having a size in the range of 5 nm to 90 nm.
1[e]. (c) a thickness of the first layer is 3 microns or less and a total coating weight of the first layer is less than 6.0 gsm; and	The EVE 50E's separator has a thickness of the first layer that is 3 microns or less and a total coating weight of the first layer is less than 6.0 gsm. As shown in the cross-section SEM image below, a thickness of the first layer is estimated to be 1.6 μm, which is less than 3 microns.



SEM image of the FIB cross-section of EVE 50E separator (annotations added).

Based on thermogravimetric analysis (TGA) of the coated separator in the EVE 50E, a total coating weight of the first layer is estimated to be 2.0 gsm, which is less than 6.0 gsm.



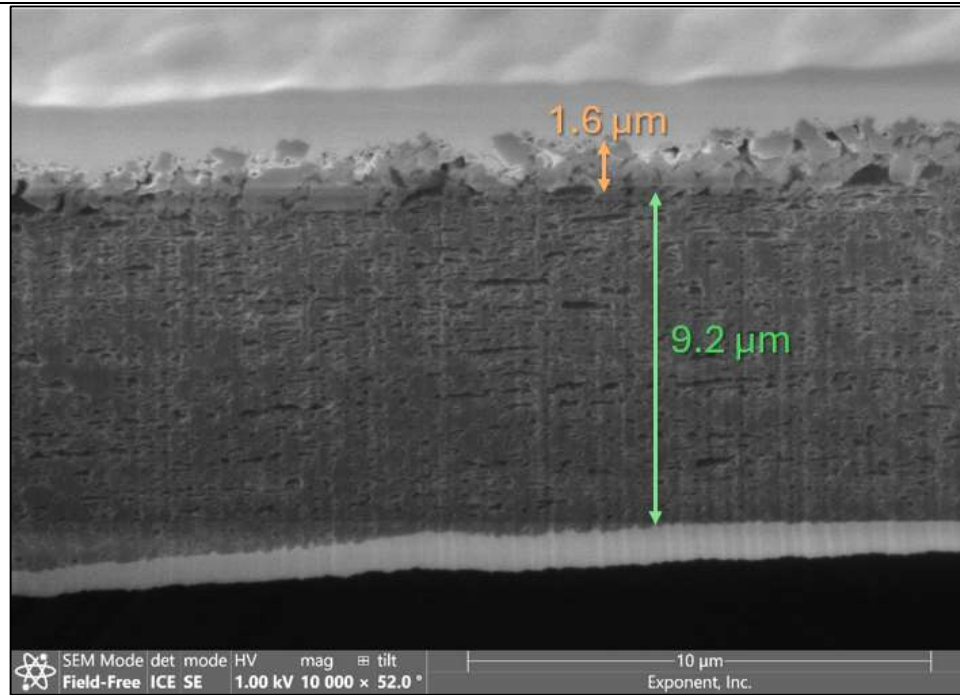
TGA analysis of the separator of the EVE 50.

Specifically, the average mass of the 8 mm diameter separator samples was 0.424 mg, which corresponds to a total separator footprint mass of 8.4 g/m². The relative mass ratio of polymeric-to-inorganic material was determined to be 76:24 based on TGA. Therefore, the coating weight of the first layer is estimated to be 8.4 g/m² × 0.24 = 2.0 g/m², which is less than 6.0 g/m². On information and belief, the presence of binder does not significantly alter this calculation.

1[f]. (d) a ratio of the thickness of the first layer to a total thickness of the separator is 10% to 80%.

The EVE 50E's separator has a ratio of the thickness of the first layer to a total thickness of the separator between 10% and 80%.

As shown below, a ratio of the thickness of the first layer (approximately 1.6 microns) to the total thickness of the separator (approximately 10.8 microns) is approximately 14.8%, which is between 10% and 80%.



SEM image of the FIB cross-section of EVE 50E separator (annotations added).